

BMU Study HUB

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FIN202 — Introduction to Financial Markets and Institutions

MOCK FINAL EXAM

Duration 90 minutes	Total Marks 100	Section A 60 marks	Section B 40 marks
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Instructions: Show ALL steps of calculation. For theoretical questions, follow Bloom's Taxonomy: start with definitions, then apply to the case, then analyze, then evaluate critically.

Calculators: Non-programmable only. Blue pen.

SECTION A — 60 Marks

4 Calculation Questions (10 marks each) + 1 Open-ended Question (20 marks)

Question 1 — Forward Contract (Cost-of-Carry Model)	10 marks
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The spot price of gold is \$1,800 per ounce. The annual risk-free interest rate is 6%. Assume no storage costs, no convenience yield, and annual compounding.

- a) Using the cost-of-carry model, calculate the one-year forward price of gold.
- b) Using the loan market and the spot market, construct a replicating portfolio for the forward contract. Demonstrate mathematically that the replicating strategy produces the same payoff as owning the forward contract.

Answer / Working:

Key Hints / Model Answer Points

Formula: $F = S \times (1 + r)^T$. Replicating: today borrow $PV = S/(1+r)$, buy gold spot; at maturity repay loan = F . Both strategies deliver the same payoff. Marking: 2pts formula setup, 3pts parameters, 3pts steps, 2pts final answer.

Question 2 — Exchange Rates & Cross Rates	10 marks
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You are given: $\text{USD/JPY} = 155$ and $\text{EUR/USD} = 1.25$.

- a) Calculate the EUR/JPY cross rate.
- b) The exchange rate moves from $\text{USD/JPY} = 155$ to 160. Calculate the percentage change and state whether the Japanese Yen (JPY) appreciated or depreciated against the USD.

Answer / Working:

Key Hints / Model Answer Points

Cross rate: $\text{EUR/JPY} = \text{EUR/USD} \times \text{USD/JPY} = 1.25 \times 155 = 193.75$. % change = $(160-155)/155 \times 100 = +3.23\%$. More JPY per USD means JPY depreciated.

Question 3 — Real Exchange Rate & PPP

10 marks

Market data (Year-End 2025):

- Spot Rate (UZS/USD) = 12,700
- Spot Rate (USD/EUR) = 1.10
- Uzbekistan CPI = 110
- Eurozone CPI = 103

- a) Calculate the UZS/EUR cross rate using the indirect method.
- b) Calculate the Real Exchange Rate (RER) for UZS/EUR using the CPI data.
- c) Last year's RER was 13,000. Has the UZS experienced real appreciation or depreciation? Critically evaluate the implications for Uzbekistan's export competitiveness.

Answer / Working:

Key Hints / Model Answer Points

$UZS/EUR = UZS/USD \times USD/EUR = 12,700 \times 1.10 = 13,970$. $RER = \text{Nominal} \times (CPI_{\text{foreign}} / CPI_{\text{domestic}}) = 13,970 \times (103/110) = \sim 13,081$. Compare to 13,000 → slight real depreciation → Uzbek exports become slightly more competitive.

Question 4 — Futures P&L**10 marks**

A trader enters a short copper futures contract at \$4.00 per pound. The contract size is 10,000 pounds.

- a) Calculate the total gain or loss if the final price is \$3.70/lb.
- b) Calculate the total gain or loss if the final price is \$4.30/lb.
- c) Briefly explain what a negative P&L on a short position tells us about the trader's initial directional view.

Answer / Working:

Key Hints / Model Answer Points

Short P&L = (Entry - Final) × contract size. (a) $(4.00 - 3.70) \times 10,000 = +\$3,000$ gain. (b) $(4.00 - 4.30) \times 10,000 = -\$3,000$ loss. A short position profits when price falls; a loss means the trader's bearish view was wrong.

Question 5 — Interest Rate Parity & No-Arbitrage Forward**10 marks**

The spot exchange rate is $GBP/USD = 1.30$. The UK interest rate is 5% per annum, and the US interest rate is 3% per annum.

- a) Calculate the one-year forward exchange rate assuming no arbitrage (Interest Rate Parity).
- b) If the actual market forward quote is $GBP/USD = 1.32$, describe the arbitrage opportunity that exists and outline the steps to exploit it.

Answer / Working:

Key Hints / Model Answer Points

$F = S \times (1 + r_{\text{USD}}) / (1 + r_{\text{GBP}}) = 1.30 \times 1.03/1.05 = 1.2752$. If market = 1.32 > 1.2752: GBP is overpriced in forward market. Arbitrage: borrow GBP at 5%, convert to USD at spot 1.30, invest USD at 3%, sell USD forward at 1.32. Lock in riskless profit.

Question 6 — Options as Insurance [OPEN-ENDED]**20 marks**

You own 5,000 shares currently worth \$25 each (total portfolio value: \$125,000). It is now May, and you are concerned about a potential decline in the share price over the next four months.

- a)** Explain how put options can be used to insure your equity position. What strike price and expiry would you choose, and why? (Refer to the cost and mechanics of the strategy.)
- b)** Describe the cash flows in full: what do you pay upfront, and what happens at expiry if (i) the share price falls to \$20, and (ii) the share price rises to \$30?
- c)** A colleague argues: 'Writing a September call option with a strike of \$20 when the stock is at \$18 (option price \$2) is a better strategy because you earn premium.' Critically evaluate this claim. Compare risk/reward with the put-buying strategy.

Answer (aim for Analyze/Evaluate level — define, apply, break down relationships, make a critical judgement):

Key Hints / Model Answer Points

(a) Buy 5,000 put options (1 per share) with strike \$25, September expiry — provides a floor. (b) Upfront cost = 5,000 x premium. If price = \$20: exercise puts, sell at \$25 → floor maintained (less premium cost). If price = \$30: puts expire worthless, keep full upside. (c) Written call earns \$2 x 5,000 = \$10,000 premium BUT provides zero downside protection and caps upside above \$20. If price surges to \$30, you lose gains above \$20. The strategies have opposite risk profiles: put-buyer pays for protection; call-writer earns premium but takes unlimited upside risk.

SECTION B — Case-Based Questions — 40 Marks

2 Case Questions x 20 marks each. Apply theory to the case. Logical flow and critical reasoning are essential.

Case Study 1 — Exchange Rate Analysis (20 marks)

Market Data (Year-End 2025):

- Spot Rate (UZS/USD) = 12,700
- Spot Rate (USD/EUR) = 1.10
- Uzbekistan CPI = 110
- Eurozone CPI = 103
- Last year's RER (UZS/EUR) = 13,000

Q1 : Calculate the UZS/EUR cross-rate using the indirect method. Show all steps.

Answer / Working:

Q2 : Compute the Real Exchange Rate (RER) for UZS/EUR using the CPI data above.

Answer / Working:

Q3: Compare the calculated RER with last year's RER of 13,000. Critically evaluate whether the UZS has experienced real appreciation or depreciation, and explain the implications for trade competitiveness.

Answer (target Evaluate level):

Key Hints / Model Answer Points

Bloom's guide: Pass = correct definition + calculation. Merit = links RER change to purchasing power. Distinction = critically evaluates direction of change, trade competitiveness implications, and considers limitations of the CPI-based RER measure.

Case Study 2 — Bond Default & Market Efficiency (20 marks)

A speculative-grade bond issuer announces, just before markets open, that it will default on its upcoming interest payment. In the days leading up to the announcement, financial media had already reported:

- Suppliers demanded cash payments, significantly reducing the issuer's liquidity.
- The issuer likely lacked sufficient cash to meet its upcoming interest payment.

Q1 — Market Efficiency:

If markets are semi-strong efficient, at what point should the bond price reflect the expected recovery value? Critically evaluate whether the price adjustment happens before, during, or after the official announcement.

Answer (target Evaluate level — 80-100%):

Q2 — Investor Behavior:

Suppose the bond price drops sharply on announcement day. Does this necessarily indicate market inefficiency? Justify your answer with reference to market microstructure.

Answer / Working:

Q3 — Strategic Implications:

You are a portfolio manager holding this bond. Describe in detail how you would respond to the default news. Reference: expected recovery value, current market price, hedging instruments, and portfolio construction principles.

Answer (target Evaluate level — aim for Distinction):

Key Hints / Model Answer Points

Q1 (Evaluate): Semi-strong EMH -> prices reflect all public info. Media reports were public -> rational investors price in default before announcement. Official statement confirms, does not introduce new info. Announcement-day volatility = liquidity shock / uncertainty resolution, consistent with efficiency.

Q2: No. Efficiency != smooth prices. Sharp drop can reflect rational reassessment, herding, or liquidity exits — all consistent with efficiency. Volatility != inefficiency. Q3 (Distinction): Compare market price vs recovery (\$0.36-\$0.38/\$1). Sell if price above recovery; hold if below. Use CDS for hedging.

Reduce speculative-grade concentration. Frame all decisions within market efficiency principles.

Marking Logic — Bloom's Taxonomy (Qualitative Questions)

Level	Skill	Marks Range	Indicators
Remember	Recall facts/definitions	40–49%	Basic description, no reasoning, no supporting points
Understand	Explain ideas	50–59%	Simple cause-and-effect link, general subject knowledge
Apply	Use theory in context	60–69%	Structured reasoning, correct logic, application to the case
Analyze	Detailed analysis	70–79%	Multiple factors, clear flow, breakdown of relationships, tied to theory
Evaluate/Create	Judge, recommend, innovate	80–100%	Critical insight, strategic thinking, justified recommendations

Good luck! Structure your answers and trust your preparation.

Prepared by BMU Study HUB

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